

# VANESHA HARI

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Available **Jan–Aug 2027** for co-op/internship in AI/ML, embedded systems, or hardware engineering. Computer Engineering student at Virginia Tech with internships at Veeam (RAG), D-Wave Quantum (PyTorch), and VT CPES (power electronics).

## EDUCATION & SKILLS

**Virginia Tech** — B.S. Computer Engineering, Minor in Computer Science | Dean's List *Graduation: May 2028*

**Languages:** Python, C++, Embedded C/C++, Java, JavaScript, HTML/CSS

**AI/ML:** PyTorch, LLMs, RAG, Hybrid & Vector Search, Query Expansion, Reranking, Embeddings, LangChain, Hugging Face, FAISS

**Embedded & Hardware:** ESP32, Arduino, Firmware Development, Interrupt-Driven Programming, Analog Signal Conditioning, Circuit Design, Power Electronics

**Tools & Platforms:** Git, Docker, SolidWorks, MATLAB, Linux

**Coursework:** Embedded Systems, Data Structures & Algorithms, Circuits & Devices, Signals & Systems, Computational Engineering, Discrete Mathematics

## WORK EXPERIENCE

**AI Engineering Intern, Veeam Software** *May – Aug 2026*

- **Improve enterprise RAG retrieval quality** by evaluating hybrid search, reranking, query expansion, and embedding strategies against production baselines.
- **Build Python experimentation frameworks** to benchmark retrieval accuracy across 5,000+ customer support cases, surfacing failure modes that informed search-relevance improvements.

**Machine Learning Developer Intern, D-Wave Quantum**

*May – Aug 2025*

- **Developed a PyTorch movie recommender** using quantum-inspired Restricted Boltzmann Machines (RBMs); evaluated model performance on user-item interaction data.

**Research Intern, Virginia Tech Center for Power Electronics Systems (CPES)**

*Jun – Aug 2023*

- **Selected as 1 of 4 students nationwide** to research nano-silver sintering for EV semiconductor packaging; work featured in Virginia Tech News [Featured in Virginia Tech News for undergraduate semiconductor research](#)

**Product & Technical Intern, Sensoria — Seattle, WA**

*May – Sep 2022*

- Researched embedded sensing and Bluetooth-enabled medical devices for remote patient monitoring.

## PROJECTS

**Oscillometric Blood Pressure Cuff** | ESP32, Embedded C/C++, Analog Circuits, Wi-Fi

- Designed end-to-end BP monitor: analog signal conditioning (INA126P instrumentation amplifiers, Sallen-Key filters), embedded peak-detection firmware, and a Wi-Fi dashboard for real-time monitoring and AI-assisted interaction.

**SummaNote Chrome Extension** | JavaScript, HTML/CSS, LLM APIs

- Built an AI-powered browser extension that auto-extracts webpage content and delivers on-demand summarization and text-to-speech.

## LEADERSHIP & EXTRACURRICULAR

**Virginia Tech BOLT — Electric Motorcycle Design Team**

*Jan 2025 – Present*

- Designed battery-pack packaging, wiring harnesses, and sensor-integrated subsystems; developed embedded C/C++ firmware on ESP32 including a custom push-button locking mechanism modeled in SolidWorks.

**CodePath — Applications of AI Engineering Fellow (Incoming)**

*Sept 2026*

- Selected for CodePath's competitive Applications of AI Engineering program.

**Co-Founder and Advisor, [Joys of Giving](#)**

*2020 – Present*

- Raised \$20K+, mobilized 15+ volunteers, engaged 1,000+ youth, and provided 1,500+ meals through community programs.

## HONORS & AWARDS

- **National:** [U.S. Congress Gold Medal](#); Presidential Volunteer Service Award (Gold); Robotics World Championship Participant (2023).
- **Industry & Community:** 3× Society of Women Engineers (SWE) Awards; 3× NCWIT Awards; [425 Magazine 30 Under 30](#).